ISLR Ch 4.7 #6, 8, 9 & 10

6.

(a)

 6a

(b)

 6a

8.

 8  8-a  8-b

10.

In [1]:

```
# Exercise 10 (a) Numerical and graphical summaries
library(ISLR)
library(MASS)
library(class)

names(Weekly)
dim(Weekly)
summary(Weekly)

# Pairwise correlations among the numeric variables (Direction is column 9)
cor(Weekly[, -9])

# Graphical summaries
pairs(Weekly)
plot(Weekly$Volume, ylab = "Volume (billions of shares)", main = "Volume over time (week
boxplot(Today ~ Direction, data = Weekly, main = "Today's return by Direction")
table(Weekly$Direction)
```

'Year' · 'Lag1' · 'Lag2' · 'Lag3' · 'Lag4' · 'Lag5' · 'Volume' · 'Today' · 'Direction'

1089 · 9

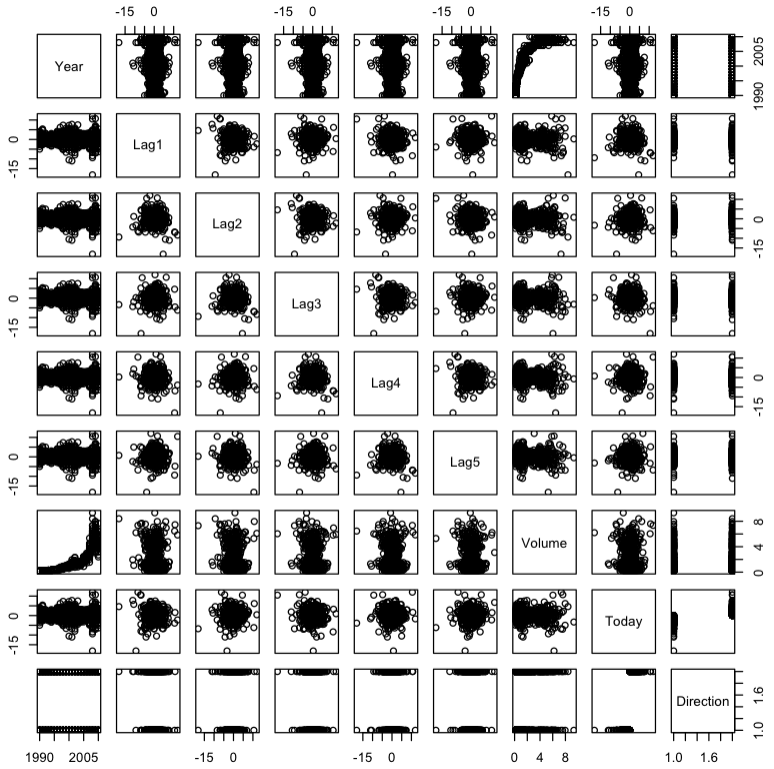
```
      Year           Lag1                Lag2                Lag3          
 Min.   :1990   Min.   :-18.1950   Min.   :-18.1950   Min.   :-18.1950  
 1st Qu.:1995   1st Qu.: -1.1540   1st Qu.: -1.1540   1st Qu.: -1.1580  
 Median :2000   Median :  0.2410   Median :  0.2410   Median :  0.2410  
 Mean   :2000   Mean   :  0.1506   Mean   :  0.1511   Mean   :  0.1472  
 3rd Qu.:2005   3rd Qu.:  1.4050   3rd Qu.:  1.4090   3rd Qu.:  1.4090  
 Max.   :2010   Max.   : 12.0260   Max.   : 12.0260   Max.   : 12.0260  
      Lag4                Lag5              Volume            Today          
 Min.   :-18.1950   Min.   :-18.1950   Min.   :0.08747   Min.   :-18.1950  
 1st Qu.: -1.1580   1st Qu.: -1.1660   1st Qu.:0.33202   1st Qu.: -1.1540  
 Median :  0.2380   Median :  0.2340   Median :1.00268   Median :  0.2410  
 Mean   :  0.1458   Mean   :  0.1399   Mean   :1.57462   Mean   :  0.1499  
 3rd Qu.:  1.4090   3rd Qu.:  1.4050   3rd Qu.:2.05373   3rd Qu.:  1.4050  
```

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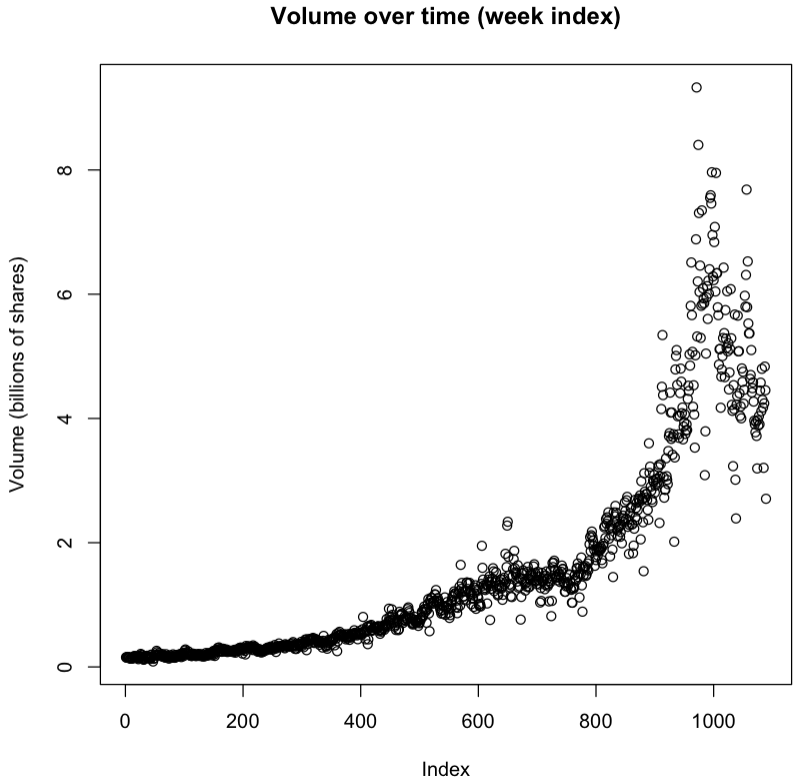
```
Max.   : 12.0260   Max.   : 12.0260   Max.   :9.32821   Max.   : 12.0260
Direction
Down:484
Up  :605
```

A matrix: 8 × 8 of type dbl

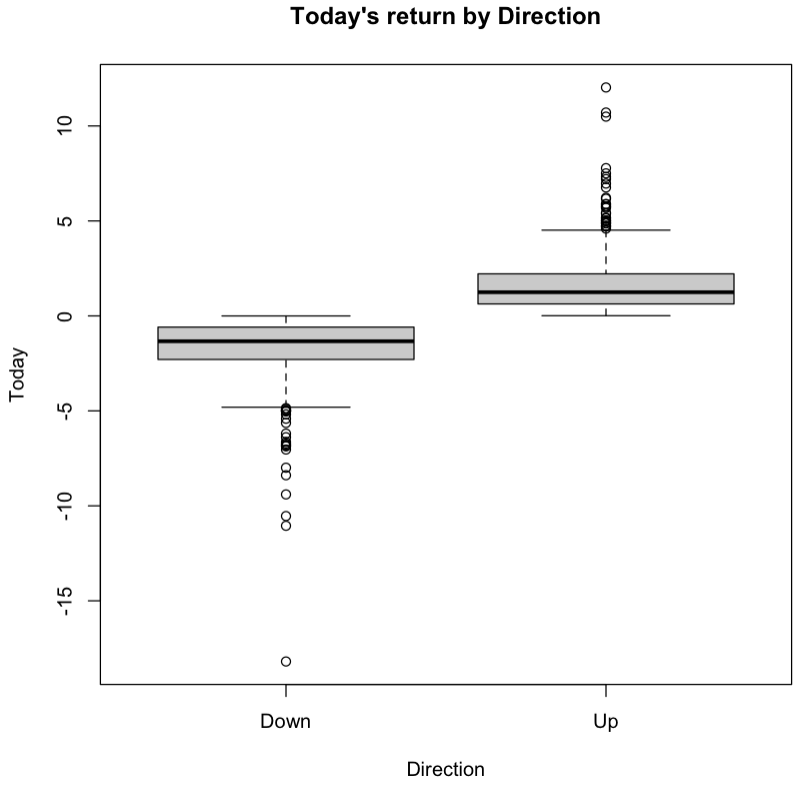
	Year	Lag1	Lag2	Lag3	Lag4	Lag5	Volume	
Year	1.00000000	-0.032289274	-0.03339001	-0.03000649	-0.031127923	-0.030519101	0.84194162	-
Lag1	-0.03228927	1.000000000	-0.07485305	0.05863568	-0.071273876	-0.008183096	-0.06495131	-
Lag2	-0.03339001	-0.074853051	1.00000000	-0.07572091	0.058381535	-0.072499482	-0.08551314	
Lag3	-0.03000649	0.058635682	-0.07572091	1.00000000	-0.075395865	0.060657175	-0.06928771	-
Lag4	-0.03112792	-0.071273876	0.05838153	-0.07539587	1.000000000	-0.075675027	-0.06107462	-
Lag5	-0.03051910	-0.008183096	-0.07249948	0.06065717	-0.075675027	1.000000000	-0.05851741	
Volume	0.84194162	-0.064951313	-0.08551314	-0.06928771	-0.061074617	-0.058517414	1.00000000	-
Today	-0.03245989	-0.075031842	0.05916672	-0.07124364	-0.007825873	0.011012698	-0.03307778	



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```
Down   Up
 484  605
```



In [2]:

```
# (b) Logistic regression on the full data set
glm.fit <- glm(Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
               data = Weekly, family = binomial)
summary(glm.fit)
```

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```
Call:
glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
    Volume, family = binomial, data = Weekly)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.26686    0.08593   3.106   0.0019 **
Lag1        -0.04127    0.02641  -1.563   0.1181
Lag2         0.05844    0.02686   2.175   0.0296 *
Lag3        -0.01606    0.02666  -0.602   0.5469
Lag4        -0.02779    0.02646  -1.050   0.2937
Lag5        -0.01447    0.02638  -0.549   0.5833
Volume      -0.02274    0.03690  -0.616   0.5377
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1496.2  on 1088  degrees of freedom
Residual deviance: 1486.4  on 1082  degrees of freedom
AIC: 1500.4

Number of Fisher Scoring iterations: 4
```

In [3]:

```r
# (c) Confusion matrix and overall fraction of correct predictions
glm.probs <- predict(glm.fit, type = "response")
glm.pred.full <- ifelse(glm.probs > 0.5, "Up", "Down")
table(glm.pred.full, Weekly$Direction)
mean(glm.pred.full == Weekly$Direction)
```

```
glm.pred.full Down  Up
         Down   54  48
         Up    430 557
```

0.561065197428834

In [4]:

```r
# (d) Logistic regression, train 1990-2008, Lag2 only; evaluate on 2009-2010
train <- Weekly$Year < 2009
Weekly.test <- Weekly[!train, ]
Direction.train <- Weekly$Direction[train]
Direction.test  <- Weekly$Direction[!train]

glm.fit2 <- glm(Direction ~ Lag2, data = Weekly, family = binomial, subset = train)
glm.probs2 <- predict(glm.fit2, Weekly.test, type = "response")
glm.pred <- ifelse(glm.probs2 > 0.5, "Up", "Down")
table(glm.pred, Direction.test)
mean(glm.pred == Direction.test)
```

```
        Direction.test
glm.pred Down Up
    Down    9  5
    Up     34 56
```

0.625

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